

Material Safety Data Sheet



QSR RESTROOM ODOR NEUTRALIZER

(In accordance with Article 10 Paragraph 1, Industrial Safety and Health Act)

Section 1. Chemical product and company identification

- a. **Korean name** : QSR 화장실 악취제거제
- Product name English** : QSR RESTROOM ODOR NEUTRALIZER
- b. **Recommended use of the chemical** : Air Freshener
- Restrictions on use** : Use only for the purpose on the product label.
- c. **Manufacturer/Supplier/Distributor** : Kay Chemical Company
8300 Capital Drive
Greensboro NC 27409
- Emergency telephone number** : 00798-14-800-8016

Section 2. Hazards identification

- a. **Hazard classification** : FLAMMABLE LIQUIDS - Category 3
- b. **GHS label elements, including precautionary statements**
- Symbol** : 
- Signal word** : Warning
- Hazard statements** : Flammable liquid and vapor.
- Precautionary statements**
- Prevention** : Wear protective gloves. Wear eye/face protection. Keep away from ignition sources such as heat/sparks/open flame. - No smoking. Use explosion-proof electrical/ventilating/lighting/material-handling equipment. Use only non-sparking tools. Take precautionary measures against static discharge. Keep container tightly closed.
- Response** : IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- Storage** : Store in cool/well-ventilated place.
- Disposal** : Dispose of contents and container in accordance with all local, regional, national and international regulations.
- c. **Other hazards which do not result in classification** : Not available.

Section 3. Composition/information on ingredients

Code : 924881-01

Ingredient name	Synonym	CAS number	%
propan-2-ol	alcool isopropilico (italian); alcool isopropylique (french); avantine; dimethylcarbinol; isohol; isopropanol; iso-propylalcohol (german); lutosol; petrohol; propropan-2-ol; 2-propanol; i-propanol (german); sec-propyl alcohol; i-propylalcohol; ipa; propan-2-ol; sec-propyl alcohol; arquad dmcb, component of (with 169153); visco 1152, component of (with 107601); pro	67-63-0	1 - 5

Section 3. Composition/information on ingredients

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in section 8.

Section 4. First aid measures

- a. **Eye contact** : In case of contact, immediately flush eyes with plenty of water. Remove contact lenses and flush again. Get medical attention if irritation persists.
- b. **Skin contact** : In case of contact, immediately flush skin with plenty of water. Wash clothing before reuse.
- c. **Inhalation** : If irritation persists, get medical attention.
- d. **Ingestion** : Do not induce vomiting. Never give anything by mouth to an unconscious person. If irritation persists, get medical attention.

e. Most important symptoms/effects, acute and delayed

Potential acute health effects

- Inhalation** : No known significant effects or critical hazards.
- Ingestion** : No known significant effects or critical hazards.
- Skin contact** : No known significant effects or critical hazards.
- Eye contact** : No known significant effects or critical hazards.

Over-exposure signs/symptoms

- Inhalation** : No specific data.
- Ingestion** : No specific data.
- Skin** : No specific data.
- Eyes** : No specific data.

f. Indication of immediate medical attention and special treatment needed, if necessary

- Specific treatments** : Not available.
- Notes to physician** : No specific treatment. Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

See toxicological information (section 11)

Section 5. Fire-fighting measures

a. Extinguishing media

- Suitable** : Use dry chemical, CO₂, water spray (fog) or foam.
- Not suitable** : Do not use water jet.

b. Specific hazards arising from the chemical : Flammable liquid and vapor. In a fire or if heated, a pressure increase will occur and the container may burst, with the risk of a subsequent explosion. Runoff to sewer may create fire or explosion hazard.

- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide

c. Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

- Special precautions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.

Section 5. Fire-fighting measures

Remark : Not available.

Section 6. Accidental release measures

- a. **Personal precautions, protective equipment and emergency procedures** : Shut off all ignition sources. Do not touch or walk through spilled material. No flares, smoking or flames in hazard area. Do not allow to enter drains or watercourses. Put on appropriate personal protective equipment (see section 8). Provide adequate ventilation.
- b. **Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).
- c. **Methods and materials for containment and cleaning up**
 - Small spill** : Stop leak if without risk. Move containers from spill area. Dilute with water and mop up if water-soluble or absorb with an inert dry material and place in an appropriate waste disposal container. Use spark-proof tools and explosion-proof equipment. Dispose of via a licensed waste disposal contractor.
 - Large spill** : Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see section 13). Use spark-proof tools and explosion-proof equipment. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Note: see section 1 for emergency contact information and section 13 for waste disposal.

Section 7. Handling and storage

- a. **Precautions for safe handling** : Put on appropriate personal protective equipment (see section 8). Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Do not ingest. Avoid contact with eyes, skin and clothing. Avoid breathing vapor or mist. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Do not enter storage areas and confined spaces unless adequately ventilated. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Store and use away from heat, sparks, open flame or any other ignition source. Use explosion-proof electrical (ventilating, lighting and material handling) equipment. Use only non-sparking tools. Take precautionary measures against electrostatic discharges. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before transferring material. Empty containers retain product residue and can be hazardous. Do not reuse container.
- b. **Conditions for safe storage, including any incompatibilities** : Store between the following temperatures: 0 to 50°C (32 to 122°F). Store in accordance with local regulations. Store in a segregated and approved area. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see section 10) and food and drink. Eliminate all ignition sources. Separate from oxidizing materials. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.

Section 8. Exposure controls/personal protection

a. Control parameters

Ingredient name	Exposure limits
propan-2-ol	Ministry of Labor (Republic of Korea, 6/2008). STEL: 980 mg/m ³ 15 minute(s). STEL: 400 ppm 15 minute(s). TWA: 480 mg/m ³ 8 hour(s). TWA: 200 ppm 8 hour(s).

Recommended monitoring procedures : If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment.

b. Appropriate engineering controls : Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.

Environmental exposure controls : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

c. Personal protective equipment

Respiratory protection : No protective equipment is needed under normal use conditions.

Eye protection : No protective equipment is needed under normal use conditions.

Hand protection : No protective equipment is needed under normal use conditions.

Skin protection : No protective equipment is needed under normal use conditions.

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.

Section 9. Physical and chemical properties

a. Appearance

Physical state : Liquid. [Liquid.]

Color : Colorless.

b. Odor : Fragrance-like.

c. Odor threshold : Not available.

d. pH : 6 to 8 [Conc. (% w/w): 100%]

e. Melting/freezing point : Not available.

f. Boiling point/boiling range : Not available.

g. Flash point : 47 °C (Closed cup)

h. Evaporation rate : Not available.

i. Flammability (solid, gas) : Not available.

j. Lower and upper explosive (flammable) limits : Not available.

k. Vapor pressure : Not available.

l. Solubility : Easily soluble in the following materials: cold water and hot water.

Section 9. Physical and chemical properties

- m. Vapor density : Not available.
- n. Relative density : 0.9 to 1
- o. Octanol/water partition coefficient : Not available.
- p. Auto-ignition temperature : Not available.
- q. Decomposition temperature : Not available.
- r. Viscosity : Not available.
- s. Molecular weight : Not applicable.

Section 10. Stability and reactivity

- a. Chemical stability : The product is stable.
- b. Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.
- c. Conditions to avoid : Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition.
- d. Incompatible materials : Reactive or incompatible with the following materials:
oxidizing materials
- e. Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information

- a. Information on the likely routes of exposure
 - Respiratory : No known significant effects or critical hazards.
 - Oral : No known significant effects or critical hazards.
 - Skin : No known significant effects or critical hazards.
 - Eyes : No known significant effects or critical hazards.
- b. Delayed and immediate effects and also chronic effects from short and long term exposure

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
propan-2-ol	LD50 Oral	Rat	5000 mg/kg	-
	LC50 Inhalation	Rat	16000 ppm	8 hours
	Gas.			

Conclusion/Summary : No known significant effects or critical hazards.

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure	Observation
Not available.					

Sensitization

Product/ingredient name	Route of exposure	Species	Result
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Not available.

Potential chronic health effects

- General** : No known significant effects or critical hazards.
- Inhalation** : No known significant effects or critical hazards.
- Ingestion** : No known significant effects or critical hazards.
- Skin contact** : No known significant effects or critical hazards.
- Eye contact** : No known significant effects or critical hazards.

Section 11. Toxicological information

Carcinogenicity : No known significant effects or critical hazards.

Mutagenicity : No known significant effects or critical hazards.

Teratogenicity : No known significant effects or critical hazards.

Developmental effects : No known significant effects or critical hazards.

Fertility effects : No known significant effects or critical hazards.

Chronic toxicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Conclusion/Summary : No known significant effects or critical hazards.

Carcinogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Conclusion/Summary : No known significant effects or critical hazards.

Mutagenicity

Product/ingredient name	Test	Experiment	Result
Not available.			

Conclusion/Summary : No known significant effects or critical hazards.

Teratogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Conclusion/Summary : No known significant effects or critical hazards.

Reproductive toxicity

Product/ingredient name	Maternal toxicity	Fertility	Development toxin	Species	Dose	Exposure
Not available.						

Conclusion/Summary : No known significant effects or critical hazards.

Specific target organ toxicity (single exposure)

Name	Target organs
	Narcotic effects

c. ATE value

Name	Route	Result

Other information : Not available.

Section 12. Ecological information

a. Aquatic and terrestrial toxicity

Environmental effects : No known significant effects or critical hazards.

Product/ingredient name	Test	Result	Species	Exposure
propan-2-ol	Behavior	Acute EC50 10000 mg/L	Fish	48 hours
	Mortality	Acute LC50 11130 mg/L	Fish	96 hours
	Mortality	Acute LC50 10400 mg/L	Fish	96 hours
	Mortality	Acute LC50 9640 mg/L	Fish	96 hours
	Mortality	Acute LC50 6550 mg/L	Fish	96 hours
	Mortality	Acute LC50	Fish	96 hours

Section 12. Ecological information

>1400 mg/L

b. Persistence/degradability

Product/ingredient name

Test

Result

Dose

Inoculum

Not available.

Product/ingredient nameAquatic half-lifePhotolysisBiodegradability

Not available.

c. Bioaccumulative potential

Product/ingredient nameLogP_{ow}BCFPotential

Not available.

d. Soil mobility : Not available.

e. Other adverse effects : No known significant effects or critical hazards.

Section 13. Disposal considerations

- a. Disposal methods : The generation of waste should be avoided or minimized wherever possible. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.
- b. Disposal precautions : Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe way. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Section 14. Transport information

IMDG Class

- a. UN number : Not regulated.
- b. Proper shipping name : -
- c. Classes : -
- d. Packing group : -
- e. Marine pollutant : Not a pollutant.
- f. Additional information : -
- Label :

Section 15. Regulatory information

a. Regulation according to ISHA

ISHA Article 37 : None of the components are listed.

ISHA Article 38 : None of the components are listed.

b. Regulation according to TCCA

TCCA Toxic chemicals : Not applicable

TCCA Observational chemicals : None of the components are listed.

TCCA Article 11 (Banned) : None of the components are listed.

TCCA Article 11 (Restricted) : None of the components are listed.

TCCA Article 14 (TRI) : The following components are listed: 2-Propanol

Korea inventory (KECI) : All components are listed or exempted.

c. Dangerous Materials Control Act : Not available.

Section 15. Regulatory information

- d. **Wastes regulation** : Dispose of contents and container in accordance with all local, regional, national and international regulations.
- e. **Regulation according to other foreign laws**
- Europe inventory** : All components are listed or exempted.
- United States inventory (TSCA 8b)** : All components are listed or exempted.
- Japan inventory (ENCS)** : All components are listed or exempted.
- Japan inventory (ISHL)** : All components are listed or exempted.

Section 16. Other information

- a. **References** : Not available.
- b. **Date of issue** : 1/27/2010.
- c. **Date of previous issue/Version** : No previous validation. / 1

☐ Indicates information that has changed from previously issued version.

Notice to reader

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